

Question number	Answer	Mark
3(a)(i)	The only correct answer is D A is not the correct answer as <i>cis retinal is changed by light</i> B is not the correct answer as <i>phytochrome is a photoreceptor in plants</i> C is not the correct answer as <i>rhodopsin is a photoreceptor found in rod cells</i>	(1)

Question number	Answer	Mark
3(a)(ii)	The only correct answer is B A is not the correct answer C is not the correct answer D is not the correct answer	(1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A description that includes two of the following points: - as light intensity increases up to 12 a.u. mean potential difference increases (1) - mean potential difference stays constant at 12 a.u. and above (1)	both mps may be in one sentence Accept maximum potential difference reached at 12 a.u.	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A calculation showing the following steps: • calculation of change in mean potential difference (1) • calculation of percentage change (1)	$20 - 17 = 3$ $3/17 \times 100 = 17.6 / 17.65 \%$ correct answer = 2 marks	2

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An answer that includes two of the following points: • removing retina (from mammals) cause {blindness/harm} (1) • using anaesthetic causes harm to mammals (1) • not ethical/ unethical (1)	do not accept killing ACCEPT benefits do not outweigh harm caused/animals cannot give consent	2

Question number	Answer	Mark
3(c)	The only correct answer is C A is not the correct answer B is not the correct answer D is not the correct answer	1